TechInsights: tracking semiconductor materials prices from public sources, reconciled, validated and documented

A reproducible pipeline and an expert Excel tracker for 15 fab materials, from raw feedstock to finished inputs, with cross-source reconciliation, root-cause analysis and nowcasts.

PUBLIC · PAGE 7

-71%

EU gallium imports in the six months after China's export licensing (Aug 2023-Jan 2024) vs the first half of 2023

PUBLIC · PAGE 6

×5.1

EU germanium import unit value, last 12 months vs 2021; polysilicon fell -53% over the same period

PUBLIC · PAGE 4

83%

of the kilograms declared as neon, krypton and xenon are implausible (about EUR 1-2/kg) and are excluded

PUBLIC · PAGE 8

2.4%

two-month nowcast error for copper using the LME price, vs 4.4% for 'last value' (24-month back-test)

QUESTIONS

1 · TechInsights What does TechInsights cover, and where do material prices fit?

2 · Coverage Which fab materials can public data track, from feedstock to wafers?

3 · Standards & quality Can sources with different units, currencies, codes and delivery terms be put on one standard and checked?

4 · Discrepancies When two sources disagree, why, and what prevents it next time?

5 · Trends & nowcast What moved prices since 2021, and can the unpublished months be estimated honestly?

Recommendations. Publish a *validated* unit value next to the raw one, with a plausible price range per material; track critical minerals *by origin*, because the origin mix moves the average more than prices do; nowcast metals from exchange prices and show intervals, not points, for thin markets; fill grade-specific gaps (electronic-grade gases, photoresists, NF_3) with a two-sided, anchor-free survey.

DATA & METHODS

PUBLIC Eurostat Comext (EU trade by CN8 code, monthly), UN Comtrade (HS6, annual), USGS Mineral Commodity Summaries 2025-2026, World Bank Pink Sheet, IMF and BLS series via FRED, ECB exchange rates; TechInsights' public website only.

SIMULATED survey answers (page 9 only), labelled as such.

Pipeline Python + DuckDB SQL, raw → clean → marts, one command: `python -m pipeline run`. 14 quality checks, 22 reconciled source pairs, 14 discrepancy cards, nowcast back-tested on 24 months.

Excel Power Query tracker with dashboard, reconciliation, FORECAST.ETS nowcast and checks.

HOW TO READ THIS REPORT

Titles state the conclusion. **Tags** PUBLIC DATA / SIMULATED show where the data comes from. **Unit value** = customs value / kg: a price *indicator* that mixes grades, not a transaction price. **Dashed lines** mark events, bands or intervals. **Red** marks alerts and discrepancies. Code, data and the Excel tracker are in the repository.

TechInsights already covers the chip from teardown to bill of materials; material input prices are the cost-side signal

PUBLIC DATA

What the company publishes about itself (techinsights.com, read on 2026-11-12) and why tracking fab-material prices fits its clients' questions.

650+	100K+	1.7M+	50K+	35+
customers	chip teardowns	images	analyses	years

COMPANY MILESTONES

1989	Founded in Ottawa as Semiconductor Insights
2016-17	Combined with Chipworks
2019-21	Acquisitions add market and manufacturing intelligence, pricing and supply chain
Materials	TECHCET is part of TechInsights: Semiconductor Materials Intelligence
Nov 2026	'Semiconductor Sustainability Outlook 2027' and 'Compute Outlook 2027' published

Content: teardowns and reverse engineering, competitive benchmarking, patent analysis, market and trend reports; free, freemium and premium tiers including **component pricing and bills of materials**; a Cost Analysis capability that breaks down the cost of materials, manufacturing and assembly.

MATERIALS SEGMENTS NAMED ON THE PUBLIC SITE

Gases · ALD/CVD precursors · Photoresists · CMP consumables · Wet chemicals · Metal plating · Sputtering targets · Equipment consumables · Critical raw materials

WHY MATERIAL PRICES MATTER TO ITS CLIENTS

- A chip's cost model needs the input side: wafers, gases, wet chemicals, metals and their **price trend**.
- Critical minerals (gallium, germanium) are now driven by **export licensing**, not only by demand: buyers need early, sourced signals.
- Clients compare figures across reports, so **one data standard and documented discrepancies** are part of the product.
- This study rebuilds that workflow with public data only, to show how I would work on it, not to replicate paid content.

THE ROLE'S TASKS, AND WHERE THIS STUDY SHOWS THEM

Task in the job description	What was built	Where
Track price trends from raw materials to finished inputs	15 materials, 4 stages	pages 2, 6, 7
Set data standards; extract, synthesise and validate data from several sources	standard schema, dictionary, code concordance	page 3, docs/DATA_STANDARDS.md
Monitor data quality; identify discrepancies, find root causes, prevent them	14 checks, 22 source pairs, 14 cards	pages 4-5, docs/DISCREPANCY_LOG.md
Document for knowledge transfer; pipelines and ETL	one-command pipeline, runbook	page 10, docs/RUNBOOK.md
Expert Excel and data extrapolation	Power Query tracker, FORECAST.ETS nowcast, back-test	page 8, excel/
Unbiased survey design (asset)	two-sided, anchor-free method, SIMULATED example	page 9, survey/

Sources: techinsights.com (About, Cost Analysis, Semiconductor Materials Intelligence, home page), himalayas.app job page; no paywalled content used · Collected 2026-11-12. Independent case study based on public data · not affiliated with or endorsed by TechInsights

All 15 fab materials have a monthly EU trade series, but only 3 have a second monthly price to check it against

PUBLIC DATA

The tracked range runs from raw feedstock to doped wafers. Coverage decides how far each series can be validated: everything else relies on annual references and checks.

RAW MATERIALS & METALS

Aluminium (unwrought) · Copper (refined cathodes) · Gallium · Germanium · Indium · Polysilicon (Si >= 99.99%) · Silicon metal (Si < 99.99%) · Tin (unwrought) · Tungsten (ores and concentrates)

GASES

Helium · Neon, krypton and xenon

CHEMICALS

Sulphuric acid · Hydrofluoric acid · Ammonia (anhydrous)

INTERMEDIATE & FINISHED

Doped silicon wafers

NO PUBLIC PRICE SOURCE

Photoresists · NF_3 · ALD/CVD precursors · CMP slurries

Hidden in broad customs codes: only a survey can price them.

SOURCE COVERAGE (BLUE = USABLE PRICE COMPARISON, GREY = PARTIAL)

ID	Material	Comext EU trade	UN Comtrade	USGS reference	Exchange price	US price index
AL	Aluminium (unwrought)	monthly	no	yes	yes	none
CU	Copper (refined cathodes)	monthly	no	yes	yes	ppi nonferrous
GA	Gallium	monthly	(HS6, shared heading)	yes	no	none
GE	Germanium	monthly	(HS6, shared heading)	yes	no	none
IN	Indium	monthly	(HS6, shared heading)	yes	no	none
POLYSI	Polysilicon (Si >= 99.99%)	monthly	(HS6)	no	no	ppi semi
SI_MET	Silicon metal (Si < 99.99%)	monthly	(HS6)	yes	no	none
SN	Tin (unwrought)	monthly	no	no	yes	ppi nonferrous
W	Tungsten (ores and concentrates)	monthly	no	content basis	no	none
HE	Helium	monthly	(HS6, shared heading)	no	no	ppi gases
NE	Neon, krypton and xenon	monthly	(HS6, shared heading)	no	no	ppi gases
H2SO4	Sulphuric acid	monthly	no	feedstock only	no	ppi inorganic
HF	Hydrofluoric acid	monthly	no	feedstock only	no	ppi inorganic
NH3	Ammonia (anhydrous)	monthly	no	yes	no	ppi inorganic
WAFER	Doped silicon wafers	monthly	(HS6)	no	no	ppi semi

One standard turns units, currencies, delivery terms and customs-code changes into comparable EUR per kg

Four raw records from four publishers, standardised by the pipeline; two customs-code splits that would break a naive time series.

BEFORE AND AFTER STANDARDISATION

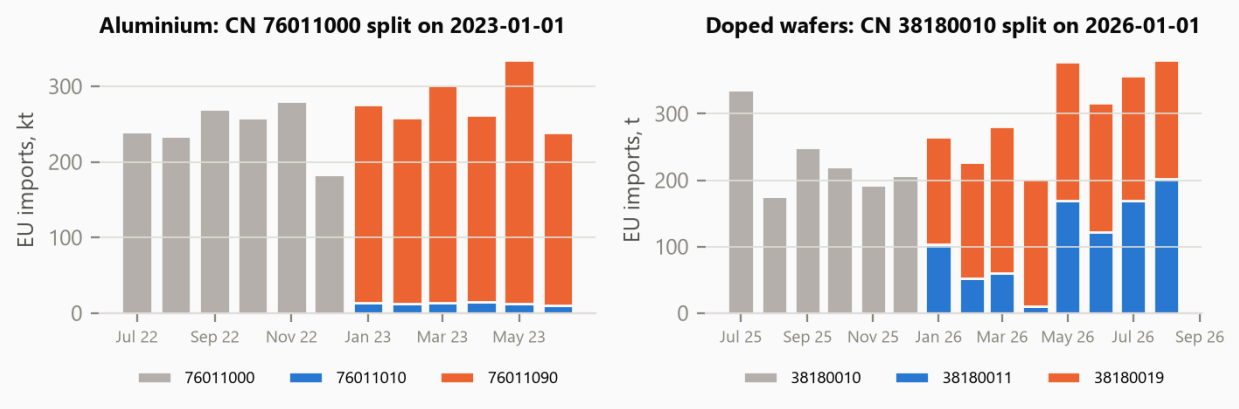
Source (as published)	Raw record	Conversion	EUR/kg	Convention
Comext, helium (CN 28042910)	15,109,748 EUR; 2,383 x 100 kg; 1,488,028 m^3; CIF	value / (100 kg × 100)	**63.41**	kg kept; m^3 used only as a check (QC08)
World Bank, copper	14,508 USD per metric ton; LME cash	/ 1,000 / 1.1202 USD per EUR	**12.95**	monthly ECB average rate
USGS, silicon metal 2025	130 US cents per pound	× 0.01 / 0.4536 / annual FX	**2.54**	US market: never mixed with EU border values
USGS, ammonia 2025	450 USD per short ton, FOB Gulf	/ 907.18 / annual FX	**0.44**	FOB vs EU CIF: compared with a band

Every source lands in one schema: `period, material_id, source, series_id, flow, reporter, partner, valuation, product_code, value_native, currency, quantity_kg, quantity_supp, price_eur_kg, price_usd_kg, collected_on`. Imports are valued **CIF** (cost, insurance, freight to the EU border), exports **FOB** (free on board); the valuation is a column, never implicit.

CONVENTIONS (DOCS/DATA_STANDARDS.MD)

- **Unit**: EUR/kg and USD/kg for every price; USGS units (cents/lb, short ton, metric ton unit) converted with documented factors.
- **Currency**: ECB monthly average; annual sources use the yearly mean.
- **Territory**: EU27_2020 for the whole history (the 'EU' total includes the UK until January 2020).
- **Codes**: a concordance table maps every CN8 code to a material with validity dates, so split codes are summed, never double counted.
- **Unit value is not a price**: it mixes grades; validated by checks and compared with references.

TWO REAL NOMENCLATURE CHANGES HANDLED BY THE CONCORDANCE



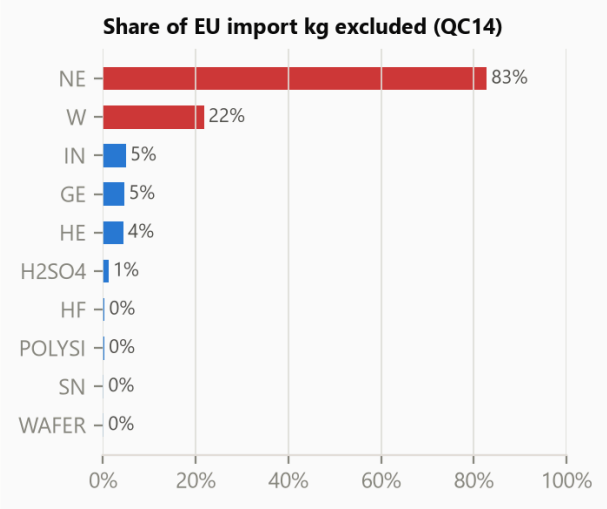
Quality checks catch what averages hide: 83% of the 'rare gas' kilograms were implausible declarations

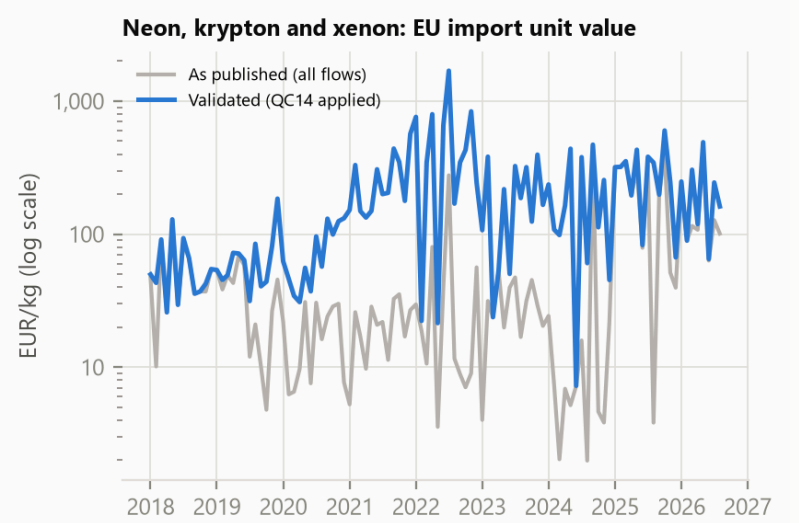
Completeness, units, outliers, nomenclature, confidentiality and freshness are tested automatically; implausible partner flows are excluded from the validated series and listed for review (docs/QUALITY_REPORT.md).

14 CHECKS, EVERY REFRESH (SQL/03_CHECKS.SQL)

Check	Rule	Severity	Rows	Materials
QC01	Missing month	error	7	2
QC02	Value without quantity	warn	6,006	15
QC03	Robust outlier (median/MAD, > 3.5 z and 25%)	warn	49	11
QC04	Thin trade month	info	24	5
QC05	Customs-code change	info	4	2
QC08	kg vs m³ inconsistent (gases)	warn	47	2
QC09	Confidential partner > 10%	info	154	8
QC11	Freshness vs normal lag	info	6	0
QC13	Comtrade quantity estimated	info	56	0
QC14	Implausible partner flow, excluded	warn	6,712	15

Also run with zero findings: codes outside validity (QC06), duplicates (QC07), withheld USGS values (QC10), missing FX (QC12).

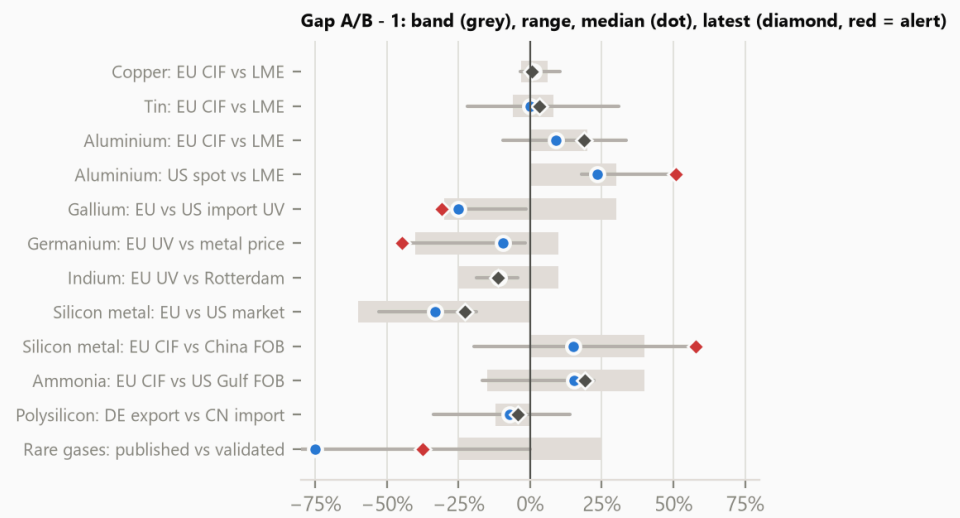


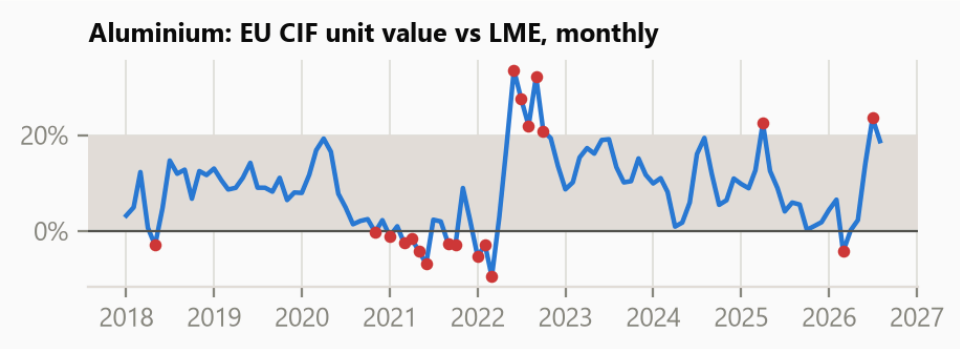


Hundreds of tonnes a year are declared under CN 28042990 at about EUR 1-2/kg, a value impossible for neon, krypton or xenon. The published average is pulled down by them; the validated series keeps only plausible flows.

Most gaps between sources have a known cause: valuation, market, definition, product mix or partner definition

22 source pairs compared; the expected band turns 'the numbers differ' into 'the difference is normal' or 'investigate'.





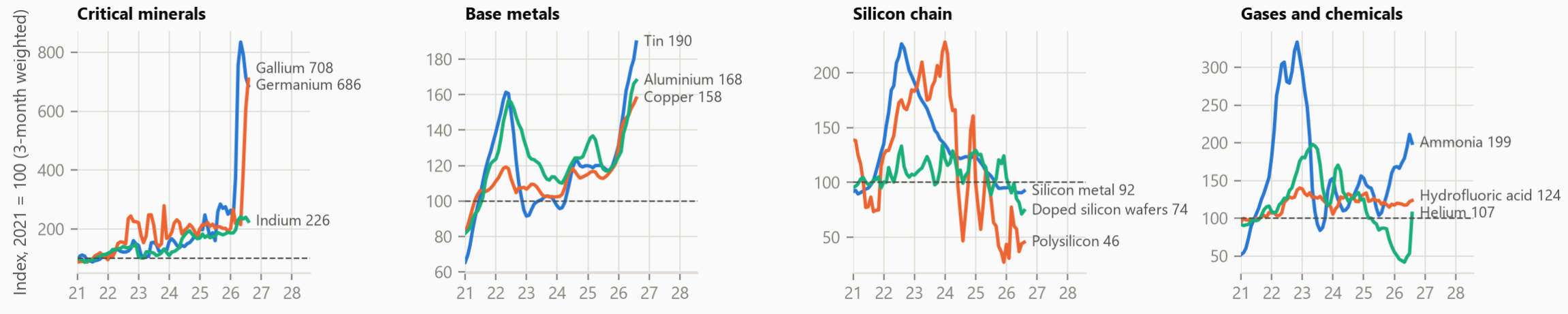
ID	Finding	Root cause	Prevention	Status
D01	**EU import unit values sit above the LME cash price**	EU border value is CIF (freight, insurance, regional premium); LME is a warehouse price.	Expected band per pair; alert only outside it.	Explained
D02	**US market prices are well above EU and LME prices**	US prices include US trade measures (Section 232 aluminium tariff, duties on silicon metal).	Market and valuation are mandatory schema fields.	Explained
D03	**Gallium 'price' fell 56% between two USGS editions for the same year**	USGS replaced a high-purity price by an import unit value between editions.	Definition text stored per edition; a change opens a new series.	Explained
D06	**One customs code holds two different products (polysilicon)**	CN 28046100 mixes US electronic-grade and Chinese solar-grade polysilicon.	Origin-stratified unit value next to the average.	Explained
D08	**Germany imports far more Chinese gallium/germanium/indium than China says it exports to Germany**	EU records imports by country of origin, China exports by destination.	Partner definition documented per source.	Explained
D11	**Published neon/krypton/xenon unit value is polluted by implausible flows**	Hundreds of tonnes declared at about EUR 2/kg: misclassified or wrong quantity.	Plausible price range per material (QC14).	Corrected

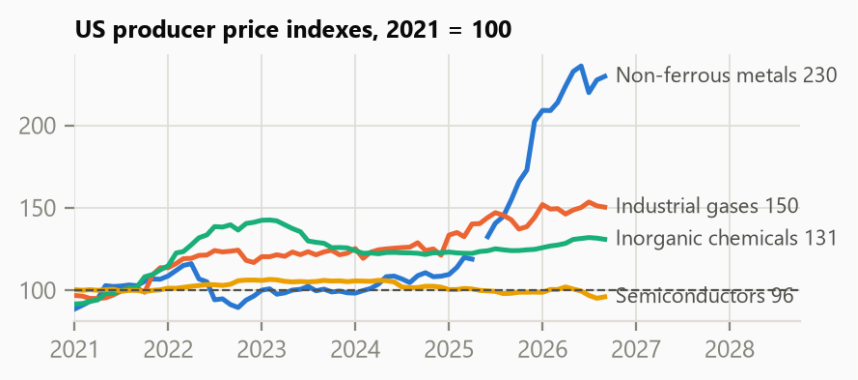
METHOD

Each pair gets an **expected band** from its known cause (e.g. CIF value 0-20% above the LME price for aluminium). Inside the band = explained; outside = alert and a card: **finding** → **root cause** → **evidence** → **correction** → **prevention**. 14 cards in docs/DISCREPANCY_LOG.md; their evidence is recomputed at each run. Two cards stay open on purpose (D07 timing, D09 wafer weights): an unexplained gap is reported, not forced into a story.

Prices diverge along the chain: critical minerals doubled to quintupled since 2021, while polysilicon, wafers and helium fell

Index of the 3-month weighted EU import unit value, 2021 average = 100, validated series. US producer price indexes give the same picture on the US side.





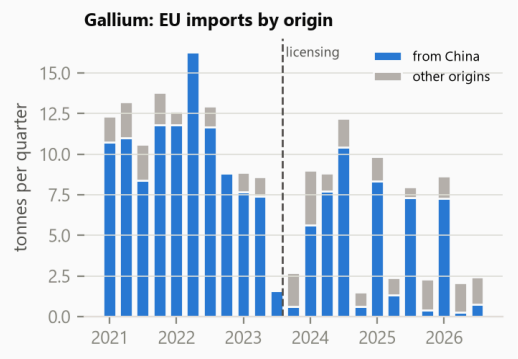
READING

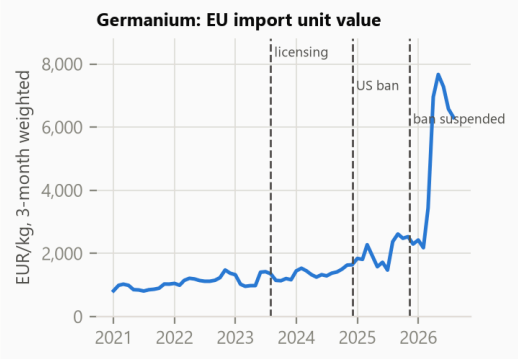
- **Critical minerals**: germanium ×5.1, gallium ×2.8, indium ×2.0 (last 12 months vs 2021, EU import unit values).
- **Base metals** follow the LME: tin +47%, copper +36%, aluminium +34%.
- **Silicon chain**: polysilicon -53% after the 2022-23 peak (and a mix shift to Chinese solar grade), wafers -8%.
- **Gases and chemicals**: helium -44% after its 2022-23 tightness; ammonia +53%, still above 2021 after the 2022 energy shock.
- Tungsten is excluded: its unit value tracks ore grade, not price (card D05).

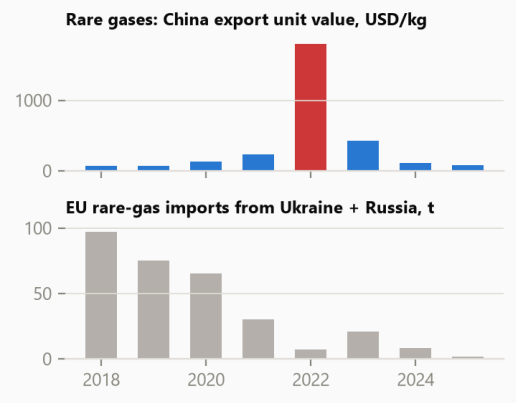
Export licensing cut EU gallium imports by 71% within six months; germanium now costs ×5.1 its 2021 level

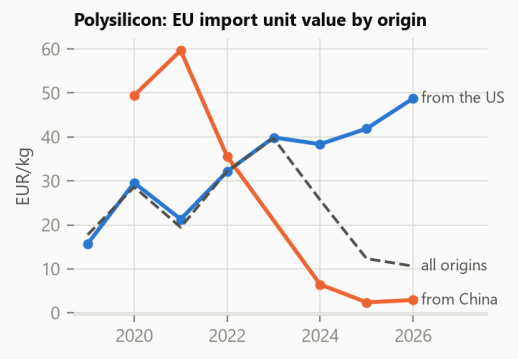
PUBLIC DATA

Events are dated from official or primary sources and described factually; charts show what EU and Chinese trade statistics recorded around them.









DATED FACTS (OFFICIAL OR PRIMARY SOURCES, SEEDS/EVENTS.CSV)

- **2023-07-03** China (MOFCOM/GAC Announcement No. 23) requires export licences for gallium and germanium items, effective **2023-08-01**.
- **2024-12-03** MOFCOM Announcement No. 46: licences for exports of gallium, germanium and antimony items to the United States in principle not approved. **2025-11-09** Announcement No. 72 suspends that US-specific ban until 2026-11-27; the general licensing regime remains.
- **2022-02/03** Ukraine's two semiconductor-grade neon producers (Cryoin, Ingas) stop production; Reuters estimated their share at 45-54% of chip-grade neon.

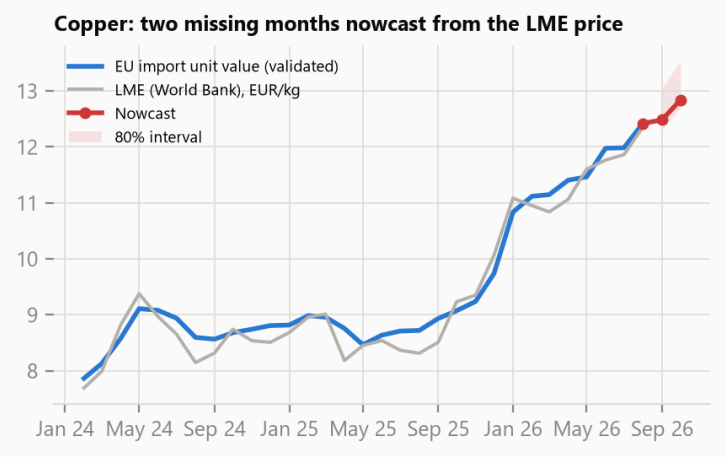
WHAT THE DATA SHOWS

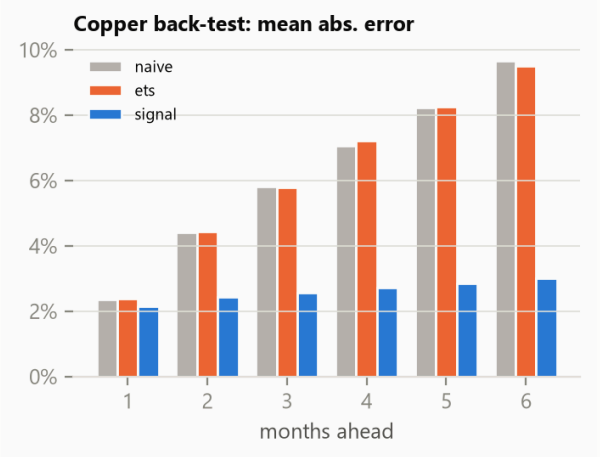
- EU gallium imports fell -71% in the six months after licensing; the share from China dropped, then partly recovered as licences were granted.
- Germanium's EU unit value rose in steps after each measure, to ×5.1 its 2021 level.
- China's rare-gas export unit value jumped in 2022 as Ukrainian and Russian supply to the EU nearly disappeared.
- Polysilicon: from 2024 cheap Chinese solar-grade metal enters the EU under the same code as US electronic grade; the all-origin average falls mostly through **mix**.

For metals, the already-published LME price cuts the two-month nowcast error from 4.4% to 2.4%; thin markets get wide, honest intervals

PUBLIC DATA

The months not yet published by Comext are estimated, tested on the past, and shown with an interval built from real past errors.





NOWCAST FOR SEPTEMBER 2026, SELECTED METHOD PER MATERIAL (EUR/KG)

ID	Method	Nowcast	80% interval	Error h1	Error h2	Naive h2
CU	signal	**12.48**	12.32 - 13.05	2.2%	2.4%	4.4%
AL	ets	**3.34**	3.20 - 3.46	2.6%	5.0%	5.2%
SI_MET	naive	**1.75**	1.65 - 1.79	3.1%	3.6%	3.6%
SN	signal	**43.04**	39.22 - 45.34	4.5%	4.4%	6.2%
HF	ets	**3.45**	3.08 - 3.67	6.5%	7.0%	8.8%
NH3	naive	**0.67**	0.62 - 0.75	7.7%	11.9%	11.9%
IN	ets	**388.99**	337.36 - 475.07	11.3%	12.1%	12.3%
WAFER	ets	**252.44**	194.19 - 297.09	15.2%	14.9%	18.5%
H2SO4	ets	**0.35**	0.31 - 0.47	16.4%	19.3%	19.6%
HE	ets	**25.39**	18.94 - 30.51	14.3%	16.7%	18.5%
GE	ets	**6,569.17**	5,973.01 - 9,572.73	22.3%	33.8%	32.5%
GA	ets	**1,065.15**	722.01 - 1,363.39	23.2%	22.2%	35.4%
POLYSI	ets	**9.96**	5.50 - 13.02	22.9%	28.6%	33.0%
W	naive	**4.55**	1.10 - 8.84	62.7%	82.0%	82.0%
NE	ets	**181.16**	46.07 - 404.66	69.1%	59.7%	82.5%

METHOD

Comext arrives about two months after the month ends; exchange prices a few days after. Three explainable methods are back-tested on 24 rolling origins ('if I had estimated N months ago'): **naive** (last value), **ets** (damped-trend exponential smoothing, the Python twin of Excel's FORECAST.ETS), **signal** (regression on the World Bank price, metals only). The method with the lowest error at one and two months is kept, and its **80% interval is the 10th-90th percentile of its own past errors**. The Excel tracker reproduces the nowcast with FORECAST.ETS and FORECAST.ETS.CONFINT.

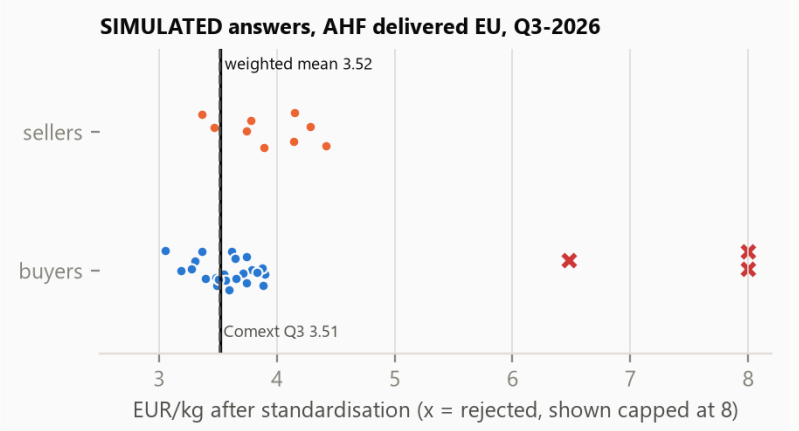
A two-sided, anchor-free survey can add the grade-specific prices that trade data cannot see

SIMULATED

Method, sample questionnaire and processing are real; the answers shown are SIMULATED (survey/process_survey.py, fixed seed) and are not TechInsights or market data.

DESIGN

- **Population**: buyers and sellers of the defined product in the EU, stratified by role and volume; large firms always included.
- **No anchoring**: no example price, no ranges to tick, no reference to last quarter; level asked before direction.
- **Definitions on the question**: purity, form, period, unit, currency, Incoterm, contract vs spot.
- **Two-sided**: buyers quote low, sellers high; both are kept apart until weighting.
- **Processing**: standardise to EUR/kg delivered → plausible range → robust outliers (called back) → design weight × capped volume → bootstrap interval.



49 invited, 73% response, 3 answers rejected (a per-tonne price typed per kg, a typo, a decimal slip). Weighted mean 3.52 EUR/kg, 90% interval 3.42-3.64; buyers 3.56 vs sellers 3.92. The match with Comext is by construction of the simulation: the point is the method.

SAMPLE QUESTION (SURVEY/QUESTIONNAIRE.MD)

*Product: anhydrous hydrogen fluoride, purity ≥ 99.9%, bulk. For deliveries during July-September 2026, what was the **average invoiced price**?*

Price: _____ (no default) · Currency: EUR / USD · Unit: per kg / per tonne · Terms: delivered (DAP/DDP) / at plant or port (EXW/FOB) · Don't know

CONSOLIDATION

The survey estimate is compared with the Comext unit value for the same quarter like any other source pair: inside the expected grade band it confirms both; outside it opens a discrepancy card.

BIAS CHECKLIST (SURVEY/METHOD.MD)

Bias	Counter-measure
Anchoring	No numbers in questions; level before direction
Strategic answers	Both sides of the market; matched panel; confidentiality (cells of 5+)
Non-response	Reminders; response rate by stratum; weights recomputed on respondents
Coverage	Frame refreshed yearly from public registers and customer lists
Definition drift	Specification printed on each question; a new specification = a new series
Unit errors	Standardisation + robust outlier test, checked with the respondent

Methods, limits and how to reproduce every number

Public sources, open code, explicit limits.

SOURCES AND LICENCES (DATA/SOURCES.MD)

Source	Content	Licence	Access
Eurostat Comext DS-045409	EU trade by CN8, monthly, value EUR, kg, supplementary units	Eurostat reuse policy (CC BY 4.0)	API, ~2-month lag
UN Comtrade	World trade by HS6, annual (public preview, no key)	UN Comtrade terms: raw records not redistributed	API
USGS MCS 2025-2026	Annual reference prices, US market	Public domain (US government)	ScienceBase
World Bank Pink Sheet	Monthly LME metal prices	CC BY 4.0	xlsx
IMF, BLS via FRED	Metal prices; producer price indexes	IMF terms / public domain	CSV
ECB	USD per EUR, monthly average	ECB reuse with attribution	API

REPRODUCE

`python -m pipeline run` rebuilds raw → clean → marts, checks, reconciliation and nowcast (no key needed); `python src/build_excel.py` and `python src/build_report.py` rebuild the tracker and this report. Monthly procedure: docs/RUNBOOK.md.

LIMITS

- A customs **unit value is not a transaction price**: it mixes grades, purities and contract terms; validated series reduce, not remove, the mix effect.
- Coverage is partial: EU border flows only; photoresists, NF_3, precursors and electronic-grade chemicals have no public price.
- Recent months are revised by Eurostat; USGS last-year values are estimates.
- Plausible ranges and expected bands are judgement calls, documented in the seeds and open to review.
- No TechInsights or TECHCET figures are used or reproduced.

AI NOTE

AI-assisted workflow: I used Claude, mainly to write scripts faster. I drove the project: I defined the questions, chose and validated the data sources, made the analytical decisions and checked every result.

PIPELINE (PIPELINE/, SQL/)

1 Extract
6 official APIs or files, polite rate, dated snapshots in data/raw

→

2 Standardise
parse, pivot, concordance; one schema in data/clean (DuckDB SQL)

→

3 Check
14 rules; implausible flows excluded; QUALITY_REPORT.md

→

4 Marts
monthly tracker, origins, annual, mirror flows, coverage

→

5 Reconcile
22 pairs, expected bands, DISCREPANCY_LOG.md

→

6 Nowcast
3 methods, 24-month back-test, 80% intervals